# Differential Geometry Excercise 2

Idan Stark

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## 1 Naturality of the Lie bracket

## 2 Verify the Jacobi identity

For every three vector fields $X$, $Y$, and $Z$:

$$\begin{gathered}
[X,[Y,Z]]+[Y,[Z,X]]+[Z,[X,Y]]= \\
= X\circ[Y,Z]-[Y,Z]\circ X+Y\circ[Z,X]-[Z,X]\circ Y+Z\circ[X,Y]-[X,Y]\circ Z= \\
= X\circ Y\circ Z-X\circ Z\circ Y-Y\circ Z\circ X+Z\circ Y\circ X+ \\
+Y\circ Z\circ X-Y\circ X\circ Z-Z\circ X\circ Y+X\circ Z\circ Y+ \\
+Z\circ X\circ Y-Z\circ Y\circ X-X\circ Y\circ Z+Y\circ X\circ Z=0
\end{gathered} \tag{1}$$